STOCHASTIC METHODS IN WATER RESOURCES

Unit 5: Geostatistics
Lecture 17: Ordinary and block kriging

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Spatial interpolation by kriging

Ordinary kriging

Ordinary kriging can be used if:

1. $Z(\mathbf{x})$ is a wide sense stationary random function but the mean of $Z(\mathbf{x})$ is unknown, or
2. $Z(\mathbf{x})$ is an intrinsic random function.

An intrinsic random function has the following properties:

$$E[Z(\mathbf{x}_2) - Z(\mathbf{x}_1)] = 0$$

$$E[Z(\mathbf{x}_2) - Z(\mathbf{x}_1)]^2 = 2\gamma(\mathbf{x}_2 - \mathbf{x}_1) = 2\gamma(\mathbf{h})$$

The mean difference between the random variates at any two locations is zero (i.e. constant mean) and the variance of this difference is a function that only depends on the separation vector $\mathbf{h}$. The function $\gamma(\mathbf{h}) = \frac{1}{2}E[Z(\mathbf{x}) - Z(\mathbf{x} + \mathbf{h})]^2$ is the semivariogram.

The ordinary kriging predictor is a weighted average of the surrounding observations

$$\hat{Z}(\mathbf{x}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{x}_i) \tag{1}$$

with $Z(\mathbf{x}_i)$ the values of $Z(\mathbf{x})$ at the observation locations (usually within a limited search neighbourhood).

Spatial interpolation by kriging

Ordinary kriging

As with the simple kriging predictor, the equation 1 must be unbiased

$$E[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = E\left[\sum_{i=1}^{n} \lambda_i Z(\mathbf{x}_i) - Z(\mathbf{x}_0)\right] = \sum_{i=1}^{n} \lambda_i E[Z(\mathbf{x}_i)] - E[Z(\mathbf{x}_0)] = 0 \quad (2)$$

As the unknown mean is constant, i.e. $E[Z(\mathbf{x}_i)] = E[Z(\mathbf{x}_0)] \forall \mathbf{x}_i, \mathbf{x}_0$, we find the following unbiasedness constraint for the λ_i

$$\sum_{i=1}^{n} \lambda_i = 1 \quad (3)$$

Apart from being unbiased we also want to have a predictor with a minimum variance of the prediction error. The error variance for predictor (see equation 1) can be written in terms of the semivariance as

$$\begin{aligned} var[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = E[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)]^2 = \\ -\sum_{i=1}^{n}\sum_{j=1}^{n} \lambda_i \lambda_j \gamma_Z(\mathbf{x}_i - \mathbf{x}_j) + 2\sum_{i=1}^{n} \lambda_i \gamma_Z(\mathbf{x}_i - \mathbf{x}_0) \end{aligned} \quad (4)$$

We want to minimize the error variance subject to the constraint given by equation 3. In other words, we want to find the set of values $\lambda_i, i = 1, 2, \cdots, n$ for which equation 4 is minimum without violating constraint in equation 3.

Spatial interpolation by kriging

Ordinary kriging

To find these, a mathematical trick is used. First the expression of the error variance is extended as follows

$$E[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)]^2 = \sum_{i=1}^{n}\sum_{j=1}^{n}\lambda_i\lambda_j\gamma_Z(\mathbf{x}_i - \mathbf{x}_j) + 2\sum_{i=1}^{n}\lambda_i\gamma_Z(\mathbf{x}_i - \mathbf{x}_0) - 2\nu\left[\sum_{i=1}^{n}\lambda_i - 1\right] \quad (5)$$

If the estimator is really unbiased, nothing has happened to the error variance as the added term is zero by definition. The dummy variable ν is called the **Lagrange multiplier**. It can be shown that if we find the set of $\lambda_i, i = 1, 2, \cdots, n$ and the value of ν for which equation 5 has its minimum value, we have also have the set of $\lambda_i(\mathbf{x}_0), i = 1, 2, \cdots, n$ for which the error variance of the ordinary kriging predictor is minimal, while at the same time $\sum \lambda_i = 1$. As with simple kriging, the minimum value is found by partial differentiation of equation 5 with respect to $\lambda_i, i = 1, 2, \cdots, n$ and ν and equating the partial derivatives to zero. This results in the following system of $(n+1)$ linear equations

$$\sum_{j=1}^{n}\lambda_j\gamma_Z(\mathbf{x}_i - \mathbf{x}_j) + \nu = \gamma_Z(\mathbf{x}_i - \mathbf{x}_0) \qquad \sum_{i=1}^{n}\lambda_i = 1 \qquad i = 1, \cdots, n \quad (6)$$

Using the Lagrange multiplier, the value for the (minimum) variance of the prediction error can be conveniently written as

$$var[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = \sum_{i=1}^{n}\lambda_i\gamma_Z(\mathbf{x}_i - \mathbf{x}_0) + \nu \quad (7)$$

Ordinary kriging

A unique solution of the system in equation 6 and a positive kriging variance is only ensured if the semivariogram function is conditionally non-negative definite. This means that for all possible $\mathbf{x}_i, \cdots, \mathbf{x}_n \in \Re(N = 1, 2 \; or \; 3)$ and for all $\lambda_1, \cdots, \lambda_n \in \Re$ such that $\sum_i \lambda_i = 1$, the following inequality must hold

$$-\sum_{i=1}^{n} \sum_{j=1}^{n} \lambda_i \lambda_j \gamma_Z(\mathbf{x}_i - \mathbf{x}_j) \geq 0 \quad (8)$$

This is ensured if one of the permissible semivariogram models in the table below is fitted to the experimental semivariogram data.

Spherical model	$\gamma(h) = \begin{cases} c\left[\frac{3}{2}\left(\frac{h}{a}\right) - \frac{1}{2}\left(\frac{h}{a}\right)^3\right], \text{ if } h \leq a \\ c, \text{ if } h > a \end{cases}$
Exponential model	$\gamma(h) = c\left[1 - e^{-h/a}\right]$
Gaussian model	$\gamma(h) = c\left[1 - e^{-h^2/a^2}\right]$
Nugget model	$\gamma(h) = \begin{cases} 0, \text{ if } h = 0 \\ 1 = , \text{ if } h > 0 \end{cases}$
Power model	$\gamma(h) = ch^{\omega}, \text{ if } 0 < \omega < 2$

where h denotes the length of the lag vector $\mathbf{h}$.

Note that the exponential and nugget models are also permissible in cases the random function is wide sense stationary. The power model, which does not reach a sill (value at which the semivariogram levels off, meaning the semivariance stops increasing with distance) can be used in case of an intrinsic random function but not in case of a wide sense stationary random function.

Spatial interpolation by kriging

Ordinary kriging

- The unknown mean μ_Z and the langrange multiplier ν require some further explanation. If all the data are used to obtain predictions at every location, at all locations the same unknown mean μ_Z is implicitly estimated by the ordinary kriging predictor.
- The Lagrange multiplier represents the additional uncertainty that is added to the kriging prediction by the fact that the mean is unknown and must be estimated.
- Therefore, if the random function is wide sense stationary, the variance of the prediction error for ordinary kriging is larger than that for simple kriging, the difference being the Lagrange multiplier.
- This can be deduced from substituting in equation 7 $\gamma_Z(\mathbf{h}) = \sigma_Z^2 - C_Z(\mathbf{h})$ and taking into account that $\sum_i \lambda_i = 1$. This means that, whenever the mean is not exactly known and has to be estimated from the data it is better to use ordinary kriging, so that the added uncertainty about the mean is taken into account.
- Even in simple kriging one rarely uses all data to obtain kriging predictions. Usually only a limited number of data close to the prediction location are used. This is to avoid that the kriging systems become too large and the mapping too slow.
- The most common way of selecting data is to center an area or volume at the prediction location $\mathbf{x}_0$. Usually the radius is taken about the size of the variogram range. A limited number of data points that fall within the search area are retained for the kriging prediction.

Ordinary kriging

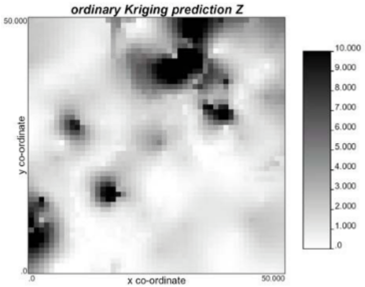
- This means that the number of data locations becomes a function of the prediction location: $n \equiv n(\mathbf{x}_0)$. Also, if ordinary kriging is used, a local mean is implicitly estimated that changes with $\mathbf{x}_0$. So we have $\mu \equiv \mu(\mathbf{x}_0)$ and $\nu \equiv \nu(\mathbf{x}_0)$ footnote .
- This shows that, apart from correcting for the uncertainty in the mean and being able to cope with a weaker form of stationarity, ordinary kriging has a third advantage when compared to simple kriging: even though the intrinsic hypothesis assumes that the mean is constant, using ordinary kriging with a search neighbourhood enables one to correct for local deviations in the mean.
- This makes the ordinary kriging predictor more robust to trends in the data than the simple kriging predictor.

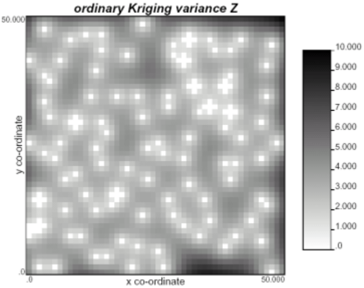
Spatial interpolation by kriging

Ordinary kriging: practical application

1. Estimate the semivariogram
2. Fit a permissible semivariogram model
3. For every node on the grid repeat:
   3.1 Solve the kriging equations: Using the fitted semivariogram model $\gamma_Z(\mathbf{h})$ in the $n+1$ linear equations 6 yields, after solving them, the kriging weights $\lambda_i, \; i = 1, 2, \cdots, n$ and the Lagrange multiplier ν.
   3.2 Predict the value $Z(\mathbf{x}_0)$: With the λ_i , the observed values $z(\mathbf{x}_i)$ (usually within the search neighbourhood) in equation 1 the unknown value of $Z(\mathbf{x}_0)$ is predicted.
   3.3 Calculate the variance of the prediction error: Using λ_i, $\gamma_z(\mathbf{h})$ and ν the variance of the prediction error is calculated with equation 7.

The figure shows the ordinary kriging prediction and the ordinary kriging variance.





Block kriging

- Up to known we have been concerned with predicting the values of attributes at the same support (averaging volume) as the observations, usually point support. However, in many cases one may be interested in the mean value of the attribute for some area or volume much larger than the support of the observations.
- For instance, one may be interested in the average porosity of a model block that is used in a numerical groundwater model, or the average precipitation of a catchment. These average quantities can be predicted using **block kriging**.
- The term block kriging is used as opposed to point kriging or punctual kriging where attributes are predicted at the same support as the observations. Any form of kriging has a point form and a block form.
- So, there is simple point kriging and simple block kriging and ordinary point kriging and ordinary block kriging. Usually, the term point is omitted and the term block is added only if the block kriging form is used.

Spatial interpolation by kriging

Block kriging

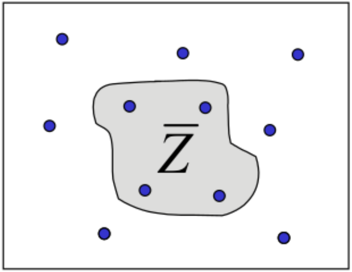
Consider the problem of predicting the mean Z of the attribute z that varies with spatial coordinate $\mathbf{x}$ for some area or volume D with size $|D|$ (length, area or volume)

$$\bar{Z} = \frac{1}{|D|} \int_{\mathbf{x} \in D} Z(\mathbf{x}) d\mathbf{x} \tag{9}$$

In case D is a block in three dimensions with upper an lower boundaries boundaries x_l, y_l, z_l, x_u, y_u, z_u the spatial integral in equation 9 stands for

$$\frac{1}{|D|} \int_{\mathbf{x} \in D} Z(\mathbf{x}) d\mathbf{x} = \frac{1}{|x_u - x_l||y_u - y_l||z_u - z_l|} \int_{z_l}^{z_u} \int_{y_l}^{y_u} \int_{x_l}^{x_u} Z(s_1, s_2, s_3) ds_1 ds_2 ds_3 \tag{10}$$

Of course, the block D can be of any form, in which case a more complicated spatial integral is used. The figure shows how to predict the mean value of Z of some irregular 2D area D.



Block kriging

Similar to point kriging, the unknown value of $\bar{Z}$ can be predicted as linear combination of the observations by assuming that the predictant and the observations are partial realizations of a random function. So, the ordinary bock kriging predictor becomes

$$\hat{\bar{Z}} = \sum_{i=1}^{n} \lambda_i Z(\mathbf{x}_i) \tag{11}$$

where the block kriging weights λ_i are determined such that $\hat{\bar{Z}}$ is unbiased and the prediction variance $var\left[\hat{\bar{Z}} - \bar{Z}\right]$ is minimal. This is achieved by solving the λ_i from the ordinary block kriging system

$$\sum_{j=1}^{n} \lambda_j \gamma_Z(\mathbf{x}_i - \mathbf{x}_j) + \nu = \gamma_Z(\mathbf{x}_i, D) \qquad \sum_{i=1}^{n} \lambda_i = 1 \qquad i = 1, 2, \cdots, n \tag{12}$$

It can be seen that the ordinary block kriging system looks almost the same as the ordinary (point) kriging system, except for the term on the right hand side which is the average semivariance between a location $\mathbf{x}_i$ and all the locations inside the area of interest D

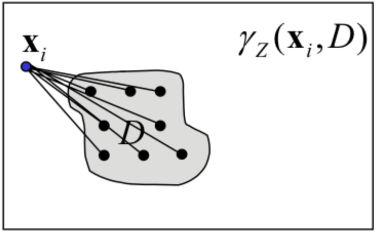
$$\gamma_Z(\mathbf{x}_i, D) = \frac{1}{|D|} \int_{\mathbf{x} \in D} \gamma_Z(\mathbf{x}_i - \mathbf{x}) d\mathbf{x} \tag{13}$$

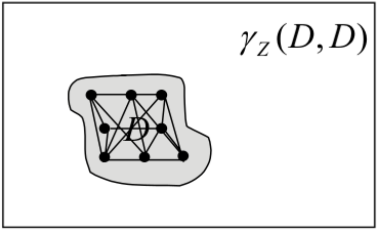
Spatial interpolation by kriging

Block kriging

When building the block kriging system, the integral in equation 13 is usually not solved. Instead, it is approximated by

1. discretizing the area of interest in a limited number of points.
2. the semivariances are calculated between the observation location and the N points $\mathbf{x}_j$ discretizing D (see left figure).





3. the average semivariance is approximated by averaging these semivariances as:

$$\gamma_Z(\mathbf{x}_i, D) \simeq \frac{1}{N} \sum_{j=1}^{N} \gamma_Z(\mathbf{x}_i - \mathbf{x}_j) \tag{14}$$

Block kriging

The variance of the prediction error is given by

$$var[\hat{\bar{Z}} - \bar{Z}] = E[\hat{\bar{Z}} - \bar{Z}]^2 = \sum_{i=1}^{n} \lambda_i \gamma_Z(\mathbf{x}_i, D) + \nu - \gamma_Z(D, D) \tag{15}$$

where $\gamma_Z(D, D)$ is the average semivariance within the area D, i.e. the average semivariance between all locations with D:

$$\gamma_Z(D, D) = \frac{1}{|D|^2} \int_{\mathbf{x}_2 \in D} \int_{\mathbf{x}_1 \in D} \gamma(\mathbf{x}_1 - \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \tag{16}$$

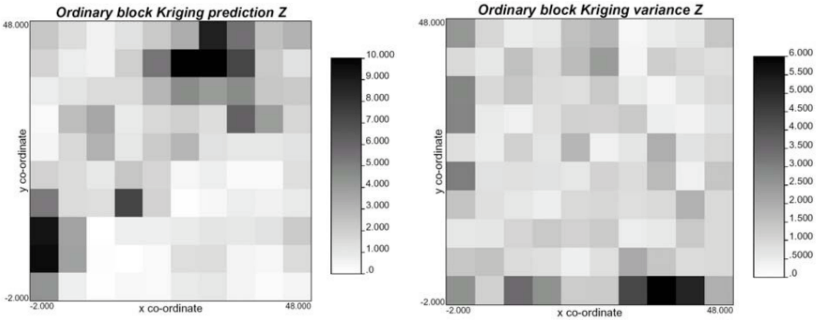
which in practice is approximated by N points $\mathbf{x}_i$ discretizing D as (see right figure above)

$$\gamma_Z(D, D) = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \gamma(\mathbf{x}_i - \mathbf{x}_j) \tag{17}$$

Spatial interpolation by kriging

Block kriging

The figure below shows the result of block kriging with block sizes of 5x5 units.



- Here we have given the equations for ordinary block kriging.
- The simple block kriging equations can be deduced in a similar manner from the simple kriging equations by replacing the covariance on the right hand side of the equation by the point-block covariance $C_Z(\mathbf{x}_i, D)$.
- In the prediction error variance, σ_Z^2 is replaced by the within block variance $C_Z(D, D)$ (the average covariance of points within D) and $C_Z(\mathbf{x}_i - \mathbf{x}_0)$ by $C_Z(\mathbf{x}_i, D)$.
- The point-block covariance and the within block covariance are defined as in equations 13 and 16 with $\gamma_Z(\mathbf{x}_1 - \mathbf{x}_2)$ replaced by $C_Z(\mathbf{x}_1 - \mathbf{x}_2)$.